

Thermal Fractional Ore Processing System

Zero-Waste, Zero-Emission Staged Thermal Separation

— v5

Overview

The Thermal Fractional Ore Processing System (TFOPS) is a zero-waste, zero-emission ore processing architecture that replaces conventional single-target extraction with staged thermal separation, capturing every mineralogical fraction as a characterized, marketable product. The system eliminates comminution energy, tailings ponds, acid mine drainage, atmospheric carbon emissions, and perpetual environmental liability. The bay architecture follows a convergent-divergent diamond topology — multiple parallel low-temperature bays feed into consolidated high-temperature stages, with LIBS-directed divergent routing downstream based on composition. Every output — from refined metal to CO₂-cured construction blocks — carries physically embedded provenance data and has a defined market.

The recommended Phase 1 configuration uses hydrogen combustion for bay heating with purchased hydrogen at approximately \$1.0 billion capital cost — less than half the capital required for a conventional mine of equivalent throughput. All process CO₂ from calcite decomposition is captured in sealed bays and mineralized into construction products. No CO₂ reaches the atmosphere. No geological storage is required. No perpetual monitoring obligations exist. The system is profitable on metals alone.

Five Design Changes

1. EM (Lorentz Force) Separation at Bay 3

Electromagnetic separation replaces the centrifuge at Bay 3. TRL 7–8, no moving parts, gravity-independent. The electromagnetic field acts on conductive liquid metals (Cu, Fe) but not on non-conductive silicate slag — five-plus orders of magnitude conductivity contrast provides highly selective separation. This eliminates the original showstopper of a TRL 3–4 centrifuge operating at 1,200°C.

2. Bay 3 at 0.5 atm (Sub-Atmospheric)

Bay 3 operates at 0.5 atm, making reducing gas escape physically impossible — any breach draws air inward, not gas outward. All off-gas is pulled to the gas management system by design. This eliminates the original showstopper of lethal reducing atmosphere containment.

3. CaO Flux at Bay 2 Before Iron Reduction

CaO flux is added at Bay 2, melting and draining silicates before iron reduction at Bay 3. This removes 80% of reactive SiO₂ before iron enters the reducing environment, structurally eliminating fayalite (Fe₂SiO₄) formation that would otherwise trap 40–70% of iron. Na₂CO₃ (27 kg/t, ~\$7/t) is added at Bay 5 to produce a standard soda-lime glass composition, resolving the original quartz viscosity showstopper.

4. Hydrogen Combustion Heating

All bays are heated by hydrogen combustion. The only combustion product is water vapor — zero CO₂, zero nitrogen oxides, zero atmospheric contaminants. This produces the cleanest possible bay atmospheres, enabling higher product purity and more accurate LIBS readings than any fossil fuel alternative. At Bay 3, hydrogen serves dual duty as both heating fuel and chemical reductant: $\text{FeO} + \text{H}_2 \rightarrow \text{Fe} + \text{H}_2\text{O}$. The reaction product is water, not CO₂, eliminating all carbon from iron reduction entirely. This is the same chemistry demonstrated at industrial scale in the HYBRIT green steel process.

5. Solar + SOFC/SOEC (Optional Phase 2)

For sites where long-term hydrogen self-production is economically justified, the energy system can be expanded to include on-site solar generation with solid oxide fuel cell and electrolyzer banks. Full agent analysis found this configuration at ~\$7.0B CAPEX is economically challenging as Day 1 but potentially attractive as a Phase 2 addition once the mine is generating revenue.

Bay Sequence

- Bay 1 (~840°C): Calcite decomposition. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$. CO_2 captured in sealed bay for sale or mineralization. CaO collected as flux for Bay 2.
- Bay 2 (~1,100°C): CaO flux added before iron reduction. Silicates melt and drain as CaO-aluminosilicate slag (mineral wool feedstock). Iron oxides and quartz remain solid. ~80% Cu recovered via heated settling vessel.
- Bay 3 (~1,200°C, 0.5 atm, H_2 reducing): Iron oxide reduced by hydrogen. $\text{FeO} + \text{H}_2 \rightarrow \text{Fe} + \text{H}_2\text{O}$. EM separation pulls conductive metals from non-conductive slag. No carbon in the system. Sub-atmospheric pressure prevents gas escape.
- Bay 4 (~1,500°C): Iron-free silicates melt. Premium iron-free glass (\$600+/t).
- Bay 5 (~1,650°C): Quartz melts with CaO + Na_2CO_3 flux. Soda-lime glass composition (83% SiO_2 , 12.5% CaO, 4.5% Na_2O). Viscosity 10–50 Pa·s.

CO2 Mineralization

TFOPS produces approximately 614,000 tonnes of process CO_2 per year from calcite decomposition at Bay 1. Because hydrogen combustion generates no combustion CO_2 , this process CO_2 is the only CO_2 in the system — captured as a pure, high-concentration stream inside a sealed bay.

- ~115,000 t/yr sold into commercial markets (food-grade, greenhouse, industrial) = ~\$16M/yr revenue
- ~400,000–500,000 t/yr mineralized into construction products. $\text{CaO} + \text{CO}_2 \rightarrow \text{CaCO}_3$ — calcium carbonate becomes the binder in CO₂-cured stone blocks, replacing Portland cement entirely.
- Zero atmospheric release. Zero geological storage. Zero perpetual monitoring.

CO₂-cured blocks gain 10–30% compressive strength improvement. The carbon is permanently sequestered in solid mineral form. Every block is a measurable, certifiable unit of permanent carbon sequestration — a carbon-negative building material.

Product Strategy

Glass and stone are zero-cost inventory — byproducts of profitable metal extraction. Any positive selling price is pure margin. Products are stockpiled indefinitely at zero holding cost during market development.

- Tier 1 — Architectural (\$5,000–50,000/t): Custom colored stone, designer glass. 1–5% of volume.
- Tier 2 — Premium construction (\$1,500–5,000/t): Iron-free glass, structural blocks, mineral wool. 5–15% of volume.
- Tier 3 — Industrial feedstock (\$100–625/t): Pre-melted glass sold to manufacturers at 25% of their retail — cheaper than their raw materials. 20–40% of volume.
- Tier 4 — Stored/backfill (\$0 carrying cost): Indefinite stockpile. Remainder.

Cast stone replaces both natural granite (zero porosity, zero fracture planes, custom color, any geometry) and laminate (\$5–15/sq ft at scale). Reconfigurable pin mold casting — an array of actuated refractory pins with a flexible refractory membrane — enables digital manufacturing: upload geometry, pour product, every piece different at commodity cost.

Integrated building system from one ore stream: stone blocks

(load-bearing), iron-free glass (translucent panels), mineral wool (insulation), cast iron (structural), CaO cement (mortar). All components carry LIBS provenance data.

Applicability Across Mining

TFOPS applies to all solid rock mining where ore is processed by crushing and separation. Eleven of fifteen most-mined metals by volume are hosted in silicate rock — directly TFOPS-compatible. Non-silicate hosts (carbonate, sulfide) also work with different product mixes. The only exceptions: brine extraction, solution mining, placer mining, coal, and dimension stone quarrying.

The lower the ore grade, the more transformative TFOPS becomes. Gangue revenue (~\$200/t) is constant regardless of metal content. A 0.3% copper ore that is uneconomic for conventional processing generates the same glass and stone revenue as a 1.0% ore. TFOPS makes low-grade deposits economic by monetizing the 99%+ that conventional mining discards.

Key Numbers

- Showstoppers remaining: 0 (all 4 original eliminated)
- Revenue streams: 12+, totaling \$278/t ore (vs \$34/t conventional)
- Phase 1 CAPEX: ~\$1.0B (H2 combustion + purchased H2)
- EBITDA: ~\$229/t ore with glass, ~\$24/t without glass — both profitable
- Tailings: zero. Acid drainage: zero. Atmospheric CO2: zero.
- Water use: 0.003 m³/t (99.7% reduction vs conventional)
- All process CO2 captured and mineralized — no geological storage
- Ore utilization: ~99.7% (vs ~1% conventional)

Space Application

The architecture is gravity and atmosphere agnostic. EM separation is gravity-independent by physics ($F = J \times B$ contains no gravity term). Hydrogen reduction chemistry works in any gravitational field. The 0.5 atm Bay 3 design translates naturally to pressure-vessel operation in vacuum — standard aerospace engineering.

The terrestrial system is not a precursor to the space system — it is the same system with environment-appropriate implementation modules installed. A company deploying terrestrial TFOPS builds direct operational experience with every subsystem the space variant requires. The technology development pathway is sequential and self-funding.

Comparison to Conventional Mining

Conventional: Exploration (cost) → Development (cost) → Operations (revenue from ~1% of ore) → Closure (cost) → Post-closure (liability)

TFOPS: Exploration (cost) → Development (cost) → Operations (12+ revenue streams from ~99.7% of ore) → Closure (revenue event) → Post-closure (ongoing lease revenue)

- Revenue: \$278/t ore vs \$34/t conventional
- CAPEX: ~\$1.0B vs \$2.5–3.5B conventional
- Ore utilization: 99.7% vs 1%
- Water: 0.003 m³/t vs 0.5–2.0 m³/t
- Tailings: zero vs 99% of ore mass
- CO₂ emissions: zero vs ~1M t/yr